

Unified Semantic Normalization and Realization Theory: Exact Realizations, Protected Obligations, and Transfer Saturation

Alp Eren Bütün

Independent Researcher, Türkiye

ORCID: 0009-0001-2479-6037

github.com/alperenbutun

September 2026

Abstract

We develop a typed exact-realization framework for deterministic semantic systems with nonunique physical realizations. Semantic evolution, optional canonicalization by sections, protected information under bounded observation, physical transport, and resource evaluation are kept as distinct layers. An exact realization is a surjective, domain-reflecting action intertwining from a deterministic physical system onto a deterministic semantic action system. Rooted path outcomes characterize transition-closed sections and give an exact fixed-template section-debt formula. Independently, protected obligations form a simplicial compatibility complex whose face-cover number yields a general capacity inequality. We prove product and transfer theorems, define category-relative transfer saturation for simultaneous compilation of independent protected distinctions, separate pairwise and global assembly defects, and quantify optimal same-model information loss. A marked-topology branch shows that holonomy has protected physical consequences only after a transport action and readout are specified. We give a source-faithful recovery audit for four source theories—completion geometry, exact pushdown normalization, chronological arithmetic, and information/resource complexity—with explicit source theorem families, typed embeddings, and reconstruction maps. As a downstream external specialization, interval-valued scalar fields supply canonical and raw common-realization obstructions of arbitrary order; the detailed boundary theorems are cited to the companion EF-FG paper rather than re-proved here. The framework is deliberately conservative: geometry, chronology, or topology alone does not imply physical cost.

Keywords. exact realization; semantic normalization; protected obligations; compatibility complexes; transfer saturation; holonomy; resource lower bounds; deterministic pushdown automata.

1 Introduction

Many deterministic systems have a simple semantic evolution and a complicated physical realization. A semantic state may admit many physical representatives; a preferred representative may fail to be transition-closed; bounded observation may be unable to distinguish futures that must remain distinct; and a topological loop may or may not act nontrivially on physical carriers. These phenomena occur in automata, normalization procedures, chronological data structures, constrained compilers, and uncertainty-realization problems, but they are governed by logically different mechanisms.

This paper develops *Unified Semantic Normalization and Realization Theory* (USNRT), a common exact-realization framework in which those mechanisms are typed separately. The central object is an exact realization

$$R = (S; C, c_0, \delta; \Theta)$$

of a rooted deterministic semantic action system

$$S = (M, m_0, \Sigma, U).$$

The projection Θ is not merely a simulation map: it is surjective on reachable states, reflects action domains, and exactly intertwines semantic and physical actions. A section σ , when present, is additional structure rather than part of semantic identity.

The main contributions are as follows.

- (i) We give a categorical exact-realization shell and distinguish representation fibers from compatibility fibers and path-outcome fibers.
- (ii) We prove a rooted endpoint factorization theorem characterizing when a transition-closed canonical section exists, and an exact formula for unavoidable fixed-template section debt.
- (iii) We introduce protected obligation complexes and prove a cover-capacity inequality converting protected incompatibility into observation/debt pressure.
- (iv) We prove product, transfer, and mixed-amplification theorems, including an exact logarithmic transfer-loss calculus.
- (v) We define category-relative transfer saturation, separate pairwise availability from global assembly, and identify exact optimal same-model loss.
- (vi) We isolate marked topology and holonomy as a separate branch: topology affects physical outcomes only through a chosen transport action, and resource conclusions require a protected readout/capacity bridge.
- (vii) We give a source-faithful four-axis recovery theorem with explicit source references, typed embeddings, reconstruction maps, and preserved theorem hypotheses.

The framework is intentionally conservative about priority. State-machine refinement mappings, refinement calculi, abstract-interpretation hierarchies, and coalgebraic system semantics already provide powerful languages for relating concrete and abstract dynamics [8, 9, 10, 11]. Rewriting theory supplies classical confluence and normal-form machinery [12, 13]; fractional covers and zero-error/confusability methods supply classical combinatorial and information-theoretic shells [14, 15, 16, 17]; and contextuality, valuation-algebra, bundle, and resource-theoretic frameworks provide established local/global and holonomy-based structures [18, 19, 20, 21]. The contribution claimed here is narrower: the typed exact-realization architecture and the transfer/non-transfer theorem package connecting optional canonicalization, protected information, chosen transport, same-model compilation, and category-specific resource evaluation.

The theory has three structural branches and one compilation layer. Branch A studies rooted path outcomes and section factorization. Branch B studies protected information through compatibility complexes and capacity. Branch C studies marked topology only after a concrete transport action has been selected. The same-model layer asks whether independently available distinctions can be compiled simultaneously inside one restricted physical category. These branches interact by theorems, not by identification. In particular, a representation fiber is not a compatibility fiber, a compatibility fiber is not a path-outcome fiber, and none of these is automatically a resource coordinate.

A companion paper develops the independent EF-FG scalar-field theory [2]. Section 13 records an exact downstream specialization of the USNRT core to that setting and cites the canonical and raw common-realization boundary theorems from the companion EF-FG paper as an external generativity test. No theorem in the present core uses those EF-FG results as premises.

2 Typed architecture and theorem scope

An enriched USNRT object may carry, in addition to the exact realization, independent geometry, chronology, observation, protected-contract, category, and resource data, which we denote schematically by

$$\mathfrak{U} = (R; \mathcal{G}_\bullet, \mathcal{J}_\bullet, \mathcal{O}_\bullet, \mathcal{B}, \text{Cat}, \varrho).$$

The notation is packaging only: the enrichments are not required simultaneously by every theorem. Geometry may be absent from a purely chronological specialization; chronology may be absent from a purely geometric one; a resource evaluation is meaningful only after a physical category has been declared.

Proposition 2.1 (Typing and non-collapse principle). *The USNRT axioms do not identify any of the following pairs by definition: semantic state and physical representative; representation fiber and compatibility fiber; section debt and behavioral failure; exact chronology and its shadow; intrinsic marked topology and physical transport; protected obligation pressure and resource cost. Any theorem transferring information between two such layers must state the required bridge data explicitly.*

Proof. The primitive data are typed independently. The exact realization contains only the semantic/physical projection and action laws. A section is optional extra structure. Geometry, chronology, observation, protected contracts, physical category, and resource evaluation are independent enrichments. The derived objects used later—path-outcome fibers, obligation complexes, holonomy orbits, and Pareto/resource bounds—are formed only after the corresponding enrichments are supplied. Therefore none of the listed identifications follows from the base definitions. $\square$

This proposition is the main scope discipline of the paper. It is deliberately negative, but it is mathematically useful: it tells us exactly which hypotheses must appear when a geometric, chronological, or topological distinction is converted into protected information or physical cost.

3 Semantic action systems and exact realizations

Definition 3.1 (Semantic action system). A rooted deterministic semantic action system is a tuple

$$S = (M, m_0, \Sigma, U),$$

where M is a set, $m_0 \in M$, Σ is an action alphabet, and each $U_a : M \rightarrow M$ is a partial function. We restrict all statements to the reachable part of M .

A semantic morphism $f : S \rightarrow S'$ preserves roots, definedness on reachable states, and actions: $f(U_a(m)) = U'_a(f(m))$ whenever $U_a(m)$ is defined.

Definition 3.2 (Exact realization). An exact realization of S is

$$R = (S; C, c_0, \delta; \Theta),$$

where C is a physical state set with root c_0 , each $\delta_a : C \rightarrow C$ is deterministic and partial, and

$$\Theta : C_{\text{reach}} \rightarrow M_{\text{reach}}$$

is surjective, root preserving, domain reflecting, and action intertwining:

$$c \in \text{Dom}(\delta_a) \iff \Theta(c) \in \text{Dom}(U_a), \quad \Theta(\delta_a(c)) = U_a(\Theta(c)).$$

An exact-realization morphism $(f, g) : R \rightarrow R'$ consists of a semantic morphism f and a rooted physical map g that intertwines physical actions and satisfies $\Theta'g = f\Theta$.

Proposition 3.3. *Exact realizations and exact-realization morphisms form a category Real_Σ .*

Proof. Identity pairs satisfy the defining conditions. If $(f, g) : R \rightarrow R'$ and $(f', g') : R' \rightarrow R''$, then roots and physical intertwining are preserved by composition and

$$\Theta''g'g = f'\Theta'g = f'f\Theta.$$

Associativity is inherited from function composition. □

Theorem 3.4 (Identity realization embedding). *The assignment*

$$I(S) = (S; M, m_0, U; \text{id}_M), \quad I(f) = (f, f)$$

defines a fully faithful functor $I : \text{SemAct}_\Sigma \rightarrow \text{Real}_\Sigma$.

Proof. Exactness is immediate for the identity projection. A morphism between identity realizations is determined by its semantic component, so the hom-set map is bijective. □

This embedding is useful conceptually: semantics is not a quotient imposed after the fact. It sits fully faithfully inside the realization category, while nontrivial realizations add physical multiplicity.

3.1 Sections are optional structure

Definition 3.5 (Sectioned realization). A sectioned realization is an exact realization together with a map

$$\sigma : M_{\text{reach}} \rightarrow C_{\text{reach}}, \quad \Theta\sigma = \text{id}.$$

When root-zero debt is required we also impose $\sigma(m_0) = c_0$.

The representation fiber is

$$\text{Fib}_{\text{rep}}(m) = \Theta^{-1}(m).$$

A section chooses one representative from each fiber, but no theorem identifies this choice with semantic identity. This distinction will matter below.

4 Path outcomes, canonicalization, and fixed-template debt

Fix a rooted exact realization. For an admissible word w , let

$$e_M(w) = U_w(m_0), \quad e_C(w) = \delta_w(c_0).$$

Define two equivalence relations on rooted admissible words by equality of semantic and physical endpoints, respectively. Exactness gives

$$e_C(u) = e_C(v) \implies e_M(u) = e_M(v),$$

but the converse may fail.

For a reachable semantic state m , define its path-outcome fiber

$$H_{c_0}(m) = \{\delta_w(c_0) : U_w(m_0) = m\}.$$

This is distinct from the full representation fiber $\Theta^{-1}(m)$: it records only physical representatives actually produced by rooted semantic paths.

Theorem 4.1 (Rooted endpoint factorization). *The following are equivalent.*

- (a) *There is a rooted transition-closed section σ through c_0 .*
- (b) *The physical endpoint map factors through the semantic endpoint map: $e_C = \sigma e_M$.*
- (c) *Semantic endpoint equivalence equals physical endpoint equivalence.*
- (d) *Every path-outcome fiber $H_{c_0}(m)$ is a singleton.*

When these conditions hold, the section is uniquely determined by c_0 .

Proof. Transition closure implies $\delta_w(c_0) = \sigma(U_w(m_0))$, giving the factorization. The factorization identifies physical endpoints whenever semantic endpoints agree; the reverse implication already follows from exactness. Equality of endpoint equivalences is precisely singleton path-outcome fibers. Finally, if each fiber is a singleton, define $\sigma(m)$ to be its unique rooted path outcome; exact intertwining makes the section transition closed. $\square$

Assume the relevant fibers are finite and define $\mu_{c_0}(m) = |H_{c_0}(m)|$. For a finite semantic test set T , set

$$\Delta_T(c_0) = \sum_{m \in T} (\mu_{c_0}(m) - 1).$$

For any rooted section σ , let $\text{Debt}_T(\sigma)$ count path outcomes over T that do not lie on the selected section.

Theorem 4.2 (Exact section-defect formula). *For every finite test region T ,*

$$\min_{\sigma} \text{Debt}_T(\sigma) = \Delta_T(c_0).$$

Proof. At a state m , a single section value can cover at most one element of $H_{c_0}(m)$, so at least $\mu_{c_0}(m) - 1$ outcomes remain off-section. Summing gives the lower bound. Choose one path outcome at each $m \in T$ as the section value and extend arbitrarily on other reachable states; then exactly one path outcome at each m lies on the section. $\square$

Corollary 4.3 (Fixed-template debt). *If $\mu_{c_0}(m) \geq 2$ for some reachable m , then no transition-closed section through c_0 exists. Refining only the observation map cannot remove this defect while the physical transition template and root representative remain fixed.*

Thus canonicalization failure is a path-factorization obstruction, not an observation-capacity statement. Observation becomes relevant only after additional protected future distinctions are specified.

5 Protected obligations and compatibility complexes

A protected obligation problem adds an observation level, confinement semantics, and a protected future contract to an exact realization. We suppress inessential data and write P_h for a finite protected problem with finite semantic source family S .

Definition 5.1 (Obligation complex). A subset $A \subseteq S$ is *one-cell compatible at level h* if the sources in A can be realized within one common level- h observation cell, under the declared confinement condition and while satisfying the protected contract. Empty sets and singletons are compatible by convention. The downward-closed family of compatible subsets is the obligation complex

$$\text{Ob}_h(P_h).$$

Its face-cover number $\kappa_h(P_h)$ is the minimum number of faces whose union covers S .

Definition 5.2 (Protected distinguishability certificate). A map $d : S \rightarrow Z$ is a protected distinguishability certificate if every face of $\text{Ob}_h(P_h)$ lies in one fiber of d .

Let D be the number of source vertices allowed to leave the declared canonical interface, X_h the number of remaining canonical sources whose obligation-realizing computation is not level- h confined, and

$$V_S(h) = |\omega_h(\sigma(S))|$$

the number of observation cells occupied by canonical confined sources.

Theorem 5.3 (Obligation-cover capacity). *If every actual level- h observation class of canonical confined sources is a face of $\text{Ob}_h(P_h)$, then*

$$D + X_h + V_S(h) \geq \kappa_h(P_h).$$

Proof. Remove the D sources permitted to leave the canonical interface and the X_h canonical sources not satisfying confinement. The remaining sources are partitioned by their actual observation cells, and each such cell is a face by hypothesis. Together with the exceptional singletons, these faces cover all source vertices, so their number is at least the minimum face-cover number. $\square$

Corollary 5.4 (Distinguishability-capacity transfer). *If $d : S \rightarrow Z$ is a protected distinguishability certificate, then*

$$\kappa_h(P_h) \geq |d(S)|, \quad D + X_h + V_S(h) \geq |d(S)|.$$

Proof. A face lies inside one fiber of d , so a face cover must contain at least one face for each represented value of d . Apply the preceding theorem. $\square$

This inequality is intentionally not a physical resource lower bound by itself. A resource conclusion requires a category-specific capacity envelope relating $D + X_h + V_S(h)$ to the resource evaluation.

5.1 Refinement and resource exclusion

Strengthening a protected contract can delete compatible faces but cannot create new ones. Hence obligation complexes are contravariantly monotone under contract refinement, while their minimum face-cover number is nondecreasing. If a physical category has a resource bound r forcing

$$D + X_h + V_S(h) \leq B_h(r),$$

then any problem with $\kappa_h(P_h) > B_h(r)$ is excluded below resource r . This is the generic route from protected incompatibility to a category-specific Pareto exclusion.

6 Products, amplification, and asymptotic obligation rate

Given finite simplicial complexes K on S and L on T , define their rectangular compatibility product $K \boxtimes L$ on $S \times T$ by declaring $A \subseteq S \times T$ a face when both coordinate projections are faces.

Theorem 6.1 (Rectangular product cover bounds). *For nonempty finite complexes,*

$$\max\{\kappa(K), \kappa(L)\} \leq \kappa(K \boxtimes L) \leq \kappa(K)\kappa(L).$$

Both bounds are sharp in general.

The upper bound comes from products of faces in optimal covers. Projecting any cover of $K \boxtimes L$ yields covers of each factor, giving the lower bound. No universal exact product law exists without additional structure.

For repeated rectangular powers $K^{\boxtimes n}$, submultiplicativity implies existence of the asymptotic rate

$$h_{\text{obl}}(K) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \kappa(K^{\boxtimes n}).$$

Let $\kappa_f(K)$ be the fractional face-cover number. The classical linear-programming fractional cover formalism [14, 17] yields exact multiplicativity under rectangular powers. A standard probabilistic rounding argument then gives the exact asymptotic identity

$$h_{\text{obl}}(K) = \log \kappa_f(K). \quad (1)$$

The combinatorial shell of (1) is classical; its role here is to quantify repeated protected-realization pressure.

The equality (1) is a statement about the derived compatibility complex. A physical interpretation requires an additional product-realization theorem.

Definition 6.2 (Iterated rectangular realization). Let P_h be a protected problem with $K = \text{Ob}_h(P_h)$. An iterated rectangular realization family is a sequence $P_h^{(n)}$ such that

$$\text{Ob}_h(P_h^{(n)}) \cong K^{\boxtimes n}$$

for every $n \geq 1$, with the declared protection and confinement semantics preserved coordinatewise.

Theorem 6.3 (Asymptotic capacity-rate exclusion). *Suppose P_h admits an iterated rectangular realization family and suppose resource bounds r_n imply certified capacity bounds*

$$D_n + X_n + V_n \leq B_n.$$

If

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log B_n < h_{\text{obl}}(K) = \log \kappa_f(K),$$

then, for all sufficiently large n , no admissible n -fold realization can satisfy resource bound r_n .

Proof. Let $\kappa_n = \kappa(K^{\boxtimes n})$. By (1), $(1/n) \log \kappa_n$ converges to $h_{\text{obl}}(K)$. Choose a number λ strictly between the two exponential rates. Eventually $\kappa_n > e^{\lambda n} > B_n$. This contradicts the one-shot obligation-capacity inequality $\kappa_n \leq D_n + X_n + V_n \leq B_n$ for any admissible realization. $\square$

Corollary 6.4 (Mixed protected rate). *If one protected problem carries two jointly independent finite certificates of ranks m and n , and their iterated physical product realization exists, then*

$$h_{\text{obl}} \geq \log m + \log n.$$

This is a protected-realization rate statement, not an identification with Shannon entropy.

7 Transfer calculus and product realization

A finite distinction transfer $\tau : X \rightarrow Y$ has logarithmic loss

$$L(\tau; X) = \log |X| - \log |\tau(X)|.$$

This measures cardinality loss only; reversibility is a stronger structured property.

Theorem 7.1 (Exact transfer-loss composition). *For a chain of effective-image-normalized transfers*

$$X_0 \xrightarrow{\tau_1} X_1 \xrightarrow{\tau_2} \dots \xrightarrow{\tau_k} X_k,$$

$$\log |X_k| = \log |X_0| - \sum_{i=1}^k L(\tau_i; X_{i-1}).$$

Equivalently, retention ratios multiply.

Proof. The logarithmic differences telescope. □

When the surviving terminal distinctions are protected, the obligation-capacity theorem converts the surviving rank into physical pressure. This gives a precise way to account for lossy bridges rather than asserting that information is “preserved” informally.

7.1 Exact product realization

Two exact realizations admit an asynchronous stutter-completed product: semantic and physical states are Cartesian products, and actions update one or both coordinates with the other coordinate stuttering. The projection and, when present, the section are coordinatewise products.

Theorem 7.2 (Exact product realization and protected product). *The asynchronous product is an exact realization. For coordinate-separable protected contracts,*

$$\text{Ob}_h(P_1 \otimes P_2) = \text{Ob}_h(P_1) \boxtimes \text{Ob}_h(P_2).$$

Proof. Surjectivity, domain reflection, and action intertwining hold coordinatewise. For protected compatibility, any product witness projects to witnesses in the factors. Conversely, factor witnesses can be run asynchronously with stutters, preserving coordinatewise confinement, observation, and protected obligations. □

Suppose the first problem has protected certificate $g : S_G \rightarrow G$ with $|g(S_G)| = m$, and the second has $j : S_J \rightarrow J$ with $|j(S_J)| = n$. Then the product certificate (g, j) has full image $g(S_G) \times j(S_J)$ and hence rank mn . Therefore

$$D + X_h + V_h \geq mn.$$

This is an abstract mixed-axis theorem. It does not assert that a restrictive one-stack pushdown category is closed under such products.

8 Same-model transfer saturation

The product theorem separates abstract coordinate products from compilation inside one restrictive physical model. Fix finite protected label sets A, B , with $|A| = m$ and $|B| = n$, and fix a declared realization category Cat together with its confinement and repair contract.

Definition 8.1 (Same-model compiler class). An object of $\text{Comp}_{\text{Cat}}(A, B)$ is a tuple

$$P = (R_P, S_P, \lambda_P, \Pi_A, \Pi_B)$$

where R_P is one realization object in Cat , S_P is a finite protected source family inside that same realization, and

$$\lambda_P : S_P \longrightarrow A \times B$$

is a joint protected-label map. The contracts Π_A and Π_B require the coordinate projections $\pi_A \lambda_P$ and $\pi_B \lambda_P$ to realize the prescribed marginal protected families under the declared confinement/repair semantics. No product closure of Cat is assumed. The joint image is

$$J_P := \lambda_P(S_P) \subseteq A \times B.$$

Define

$$\text{Sat}_{\text{Cat}}(A, B) = \max_{P \in \text{Comp}_{\text{Cat}}(A, B)} |J_P|,$$

when the class is nonempty and the finite maximum is attained. The normalized saturation ratio is $\text{Sat}_{\text{Cat}}/(mn)$, and the logarithmic loss of one compiler is $\log(mn/|J_P|)$. Thus the optimal same-model loss is

$$L_{\text{Cat}}^* = \log(mn) - \log \text{Sat}_{\text{Cat}}(A, B).$$

Proposition 8.2 (Full-saturation criterion). *The following are equivalent:*

- (a) $\text{Sat}_{\text{Cat}}(A, B) = mn$;
- (b) *the optimal loss is zero*;
- (c) *the category contains a marginal-preserving compiler whose joint image is all of $A \times B$.*

Product closure is sufficient but not necessary: a category may contain a bespoke full-rank compiler for one pair without admitting arbitrary Cartesian products.

8.1 Pairwise defect versus assembly defect

Let

$$E_{\text{Cat}}(A, B) = \bigcup_{P \in \text{Comp}_{\text{Cat}}(A, B)} J_P$$

be the pairwise envelope and $p_{\text{Cat}} = |E_{\text{Cat}}(A, B)|$. Define

$$\delta_{\text{pair}} = mn - p_{\text{Cat}}, \quad \delta_{\text{asm}} = p_{\text{Cat}} - \text{Sat}_{\text{Cat}}(A, B).$$

Proposition 8.3 (Saturation-defect identity).

$$mn - \text{Sat}_{\text{Cat}}(A, B) = \delta_{\text{pair}} + \delta_{\text{asm}}.$$

Both defects are nonnegative, and full saturation occurs exactly when both vanish.

Proof. By definition,

$$mn - \text{Sat}_{\text{Cat}} = (mn - p_{\text{Cat}}) + (p_{\text{Cat}} - \text{Sat}_{\text{Cat}}).$$

The envelope contains every joint image, so $p_{\text{Cat}} \geq \text{Sat}_{\text{Cat}}$; also $p_{\text{Cat}} \leq mn$. Hence both defects are nonnegative. Their simultaneous vanishing is equivalent to $p_{\text{Cat}} = \text{Sat}_{\text{Cat}} = mn$. $\square$

The interpretation is important. A positive pairwise defect means that some label pairs are never jointly realizable. A positive assembly defect means that every pair may occur somewhere, while no single physical realization assembles all available pairs simultaneously. Thus pairwise compatibility and simultaneous compilation are different questions even when they use the same marginal labels.

Proposition 8.4 (Elementary saturation bounds). *Whenever the compiler class is nonempty and every compiler is marginal-surjective,*

$$\max\{m, n\} \leq \text{Sat}_{\text{Cat}}(A, B) \leq mn.$$

Moreover, if $\text{Cat}_1 \subseteq \text{Cat}_2$ with the same protected witness interpretation, then

$$\text{Sat}_{\text{Cat}_1}(A, B) \leq \text{Sat}_{\text{Cat}_2}(A, B).$$

Proof. For any marginal-surjective joint image $J \subseteq A \times B$, both coordinate projections are onto, so $|J| \geq m$ and $|J| \geq n$. The upper bound is $|J| \leq |A \times B| = mn$. Enlarging the admissible compiler class can only enlarge the set over which the maximum is taken. $\square$

8.2 Shared-interface bounds

If every compiler factors its joint certificate through an interface of capacity at most C , then

$$\text{Sat}_{\text{Cat}}(A, B) \leq C.$$

Hence $C < mn$ forces positive same-model information loss, with

$$L_{\text{Cat}}^* \geq \log \frac{mn}{C}.$$

More generally, if a resource bound r gives a capacity envelope $B_h(r)$, then the resource-bounded saturation profile obeys

$$\text{Sat}_{\text{Cat}}(r; A, B) \leq B_h(r).$$

Proposition 8.5 (Optimal same-model transfer loss). *The minimum logarithmic loss among marginal-preserving compilers is exactly*

$$L_{\text{Cat}}^* = \log \frac{mn}{\text{Sat}_{\text{Cat}}(A, B)}.$$

An independent capacity bound $\text{Sat}_{\text{Cat}} \leq C$ gives a lower bound $L_{\text{Cat}}^ \geq \log(mn/C)$; an explicit compiler of joint rank R gives $L_{\text{Cat}}^* \leq \log(mn/R)$. Matching bounds certify exact loss optimality.*

Proof. Every admissible compiler has joint rank $R \leq \text{Sat}_{\text{Cat}}$ and hence loss at least

$$\log \frac{mn}{\text{Sat}_{\text{Cat}}}.$$

By definition of saturation, some admissible compiler attains rank Sat_{Cat} whenever the finite maximum is attained, and its loss is exactly the displayed value. The capacity and construction estimates follow monotonically from the same formula. $\square$

8.3 A source-stack capacity specialization

The abstract theory does not prescribe a stack or memory measure. The following numerical bound is imported from the separately stated pushdown-exposure source theorem, not derived from bare USNRT semantics. In *Collapse-Retained Information and Representation-Independent Pushdown Exposure Complexity* [7], Theorem 2.4 proves

$$D + X_h + V(h) \geq N$$

for a collapse family of size N under hidden-tail noncoalescence. Theorem 3.3 of the same paper (Source-Stack Exposure Profile) proves the deterministic top-access ε -pushdown specialization directly:

$$D + X_h + q_{\text{src}} \sum_{j=0}^h g_{\text{src}}^j \geq N.$$

Its proof derives this inequality from Theorem 2.4 by verifying hidden-tail noncoalescence and counting depth- h source signatures.

Choose one protected source representative for each distinct joint label in a same-stack compiler of joint rank R ; the resulting collapse family has size $N = R$. Applying source Theorem 3.3 gives the necessary inequality

$$R \leq D + X_h + q_{\text{src}} \sum_{j=0}^h g_{\text{src}}^j.$$

Consequently, a zero-debt, h -confined compiler with full joint rank mn must satisfy

$$q_{\text{src}} \sum_{j=0}^h g_{\text{src}}^j \geq mn.$$

For $g_{\text{src}} \geq 2$, this yields

$$h \geq \left\lceil \log_{g_{\text{src}}} \left(1 + \frac{(g_{\text{src}} - 1)mn}{q_{\text{src}}} \right) \right\rceil - 1.$$

Remark 8.6 (Necessary, not sufficient). The cited exposure theorem is a lower-bound/capacity statement. It neither constructs a one-stack compiler meeting the bound nor asserts full rectangular independence for any particular pair of geometric and chronological witnesses. Those are category-specific construction questions.

9 Marked topology, transport, and holonomy

Intrinsic topology does not itself specify a physical action. Let a marked displacement system be

$$\mathfrak{M} = (G, L, \partial),$$

where G is a loop group, L is a displacement group, and $\partial : G \rightarrow L$ is a homomorphism. The pure sector is $K_0 = \ker \partial$. A chosen physical transport representation is a homomorphism $\rho : G \rightarrow H$.

Theorem 9.1 (Pure-holonomy factorization). *The following are equivalent:*

- (a) $\rho(K_0) = \{e\}$;
- (b) $K_0 \subseteq \ker \rho$;

(c) there is a unique homomorphism $\bar{\rho} : \text{im}(\partial) \rightarrow H$ such that $\rho = \bar{\rho} \partial$.

Now suppose loops act on a physical carrier set Ω , and fix $c \in \Omega$. The orbit $A = G \cdot c$ is a set of physical path outcomes with a common semantic endpoint.

Theorem 9.2 (Orbit-to-section defect). *If the orbit A is finite, then the path-outcome multiplicity at the common semantic state is at least $|A|$, and every section of the fixed realization incurs at least $|A| - 1$ off-section outcomes there. If the action is transitive on the orbit,*

$$|A| = [G : \text{Stab}_G(c)].$$

A resource conclusion requires another bridge.

Theorem 9.3 (Protected holonomy-orbit transfer). *Assume the orbit alternatives carry a protected readout $\beta : A \rightarrow Y$ whose distinct values cannot share one level- h observation cell under the declared confined repair contract. Then*

$$D + X_h + V_A(h) \geq |\beta(A)|.$$

If β is injective, the right side is $|A|$.

Thus the valid chain is

$$\begin{aligned} \text{marked topology} &\longrightarrow \text{chosen transport} \longrightarrow \text{orbit/holonomy} \\ &\longrightarrow \text{protected readout} \longrightarrow \text{obligation pressure} \\ &\longrightarrow \text{resource, if a capacity law is supplied.} \end{aligned}$$

Deleting any of the middle hypotheses is not justified in general.

9.1 A typed transfer theorem

The previous branches can be summarized without collapsing them into one implication.

Theorem 9.4 (Typed transfer chain). *Let X be a finite structural witness family. Suppose a declared bridge produces a finite protected label image Y , and suppose every level- h observation class of confined canonical sources lies in one protected-compatible face. Then*

$$D + X_h + V_h \geq |Y|.$$

If, in addition, a category-specific resource envelope gives

$$D + X_h + V_h \leq B_h(r)$$

for all realizations of resource at most r , then $|Y| > B_h(r)$ excludes resource r . The conclusion depends on the witness-to-label bridge and on the resource envelope; neither is a consequence of the bare structural witness alone.

Proof. The bridge produces a protected distinguishability certificate on Y , so the obligation-cover theorem gives the first inequality. The second statement is immediate by contradiction from the category-specific upper envelope. $\square$

Three common instances are now transparent. A geometric compatibility witness requires a protected realization/readout. An exact chronological distinction requires a protected decoder or future obligation; a common shadow alone is insufficient. A holonomy class requires a physical transport action, then an orbit, then a protected readout. In all three cases the last step to a physical resource is category-sensitive.

10 Recovery of four source theories

USNRT was extracted from four previously developed theorem packages. The recovery claim in this section is therefore not that USNRT re-proves those source theorems from weaker axioms. It is a source-faithfulness statement: the source data, hypotheses, and theorem-bearing distinctions embed into typed USNRT components and are recovered by an explicit forgetful map. The source results used in this audit are public or separately stated manuscripts and are cited below.

10.1 Recovery datum

For a source class Src_i , let USNRT_i be the typed specialization carrying exactly the source structures listed below. A recovery datum consists of

$$I_i : \text{Src}_i \longrightarrow \text{USNRT}_i, \quad R_i : \text{im}(I_i) \longrightarrow \text{Src}_i$$

with $R_i I_i \simeq \text{id}$ up to the native source isomorphism. In addition, for each audited source theorem we record the source statement, the USNRT component that carries each hypothesis, and the reconstruction that returns the original theorem.

10.2 Axis I: completion geometry

Source. The source package is the periodic shortest-path compatibility theory in *Universal Row Complexity of Simplicial Complexes* and *Universal k -Fibers of Optimal Compatibility* [3, 4]. The audited source statements are the Exact Universal Row Complexity and Zero Shell Defect theorem (Theorem 2.2 of the former paper), the Universal k -Fiber Theorem, and the extension-fiber completeness consequences.

USNRT data and bridge. The fixed Bellman/completion system, exact distance, lower-order compatibility signature, higher-order extension fibers, and optional marked data are copied into the geometric enrichment $\mathcal{G}_\bullet$. Purely geometric instances use the identity realization; no physical resource is inferred until a separate protected-readout/capacity bridge is supplied.

Reconstruction. The map R_I forgets unused realization/protection layers and returns the original periodic system, exact distance, compatibility hierarchy, and extension fibers. In particular a universal k -fiber is not identified with a representation fiber. The Bellman coefficient construction and common-shell hypotheses remain source-specific.

10.3 Axis II: exact pushdown normalization

Source. The source package is the exact-normalization/general-theory part of the monograph [1], together with the normalization-choice specialization in *Confluence and Normalization Choice in Exact Pushdown Lifts* [5]. The audited theorem family comprises finite-observation/common-target collapse capacity, the bounded-local three-stage obstruction decomposition, and rooted path-outcome coherence.

USNRT data and bridge. Semantic dynamics, physical configurations, the exact projection, a chosen section when present, bounded observations, protected future obligations, and local schedule/action data are carried literally by the exact-realization and protection layers. Under the source tail-injectivity/noncoalescence and transition-closure hypotheses, the same-target predecessor family becomes a complete-conflict protected-obligation instance, so the USNRT capacity inequality returns the source finite-observation bound. The later allocation and local-alignment stages remain source-specific rather than consequences of the cover bound.

Reconstruction. The forgetful map returns the original semantic/physical policy, section, observation map, and local action data. It preserves the distinctions “representation debt versus behavioral correctness”, “observation collision versus hidden-tail noncoalescence”, and “capacity screen versus exact local alignment”.

10.4 Axis III: chronological arithmetic

Source. The source is *Higher-Order Chronological Carry: Connected Interval Blocks, Hessenberg Transport, and Stirling Shadows* [6]; the same theorem family is publicly archived in Chapter 34 of the research monograph [1]. The audited package consists of terminal-block separation, higher-band lower-Hessenberg realization, the subdiagonal light-cone theorem, normalized forward transport, inverse source reconstruction, the stationary parameter lattice, and the universal raw Stirling shadow.

USNRT data and bridge. The deterministic chronology is placed in $\mathcal{J}_\bullet$: exact connected source sequences, Hessenberg bands, normalized transport, inverse reconstruction, stationary parameters, and raw shadow maps remain separate typed coordinates. A common shadow is not itself a protected-information conclusion; a protected decoder/readout is an additional bridge.

Reconstruction. Forgetting unrelated realization layers returns the exact source sequences and shadow maps separately. Thus exact chronological data are not identified with their mod- p shadow, and the CCR- k theorem retains the source hypotheses under which its parameter lattice and shadow statements are valid.

10.5 Axis IV: information and resource complexity

Source. The source is *Collapse-Retained Information and Representation-Independent Pushdown Exposure Complexity* [7]. Its Theorem 2.4 gives the filtered collapse-capacity inequality

$$D + X_h + V(h) \geq N$$

under hidden-tail noncoalescence, and its Theorem 3.3 (Source-Stack Exposure Profile) gives the deterministic top-access pushdown specialization

$$D + X_h + q_{\text{src}} \sum_{j=0}^h g_{\text{src}}^j \geq N.$$

USNRT data and bridge. The source family, protected contract, observation filtration, confinement predicate, and debt/escape counts are represented by the protected-obligation branch. In the same-stack saturation specialization, choose one protected source representative for each distinct joint label. Then the collapse-family size is $N = R$, where R is the compiler’s joint rank, and source Theorem 3.3 gives exactly the bound used in Section 8.

Reconstruction. The map returns the source family, observation filtration, exposure depths, debt counts, control/alphabet parameters, and protected recoding contract. Source exposure is not identified with maximum stack height, and a capacity inequality is not identified with a matching compiler.

Theorem 10.1 (Source-faithful four-axis recovery). *For the four cited source packages above, the displayed interpretations define recovery data (I_i, R_i) satisfying*

$$R_i I_i \simeq \text{id}_{\text{Src}_i}.$$

For each audited theorem family, every source hypothesis is assigned to an explicit USNRT component or to an explicitly retained source-specific enrichment, and reconstruction returns the source theorem without adding a converse.

Proof. The four interpretations are componentwise injections of the source data listed above. In each case R_i is the projection onto those components, so $R_i I_i$ is the identity up to the source’s native renaming/isomorphism. The theorem-level claim is checked axis by axis: Bellman/common-shell data are retained on Axis I; tail noncoalescence, transition closure, and local alignment are retained on Axis II; exact Hessenberg transport and shadow maps remain distinct on Axis III; and the observation filtration plus top-access capacity law are retained on Axis IV. Since no source-specific hypothesis is deleted or promoted to a generic USNRT axiom, reconstruction yields the original implication and no additional converse. $\square$

10.6 Recovery/transfer consistency

The recovery maps also preserve the direction of the cross-axis bridges used in this paper. Four checks are especially important.

- (R1) *Axis I to Axis IV.* A universal compatibility fiber becomes a physical lower bound only after a protected collapse/readout bridge and the Axis-IV capacity theorem are inserted.
- (R2) *Axis II to obligation capacity.* Common-target finite-observation collapse is the complete-conflict specialization of protected-obligation capacity; the subsequent allocation and local-alignment stages remain source-specific.
- (R3) *Axis III to protected information.* Exact chronological alternatives with one common raw shadow create protected pressure only when a future decoder/readout distinguishes those exact alternatives.
- (R4) *Axis I topology to transport/resource.* Marked topology is distinct from a chosen physical action; resource consequences additionally require protected readout and a category-specific capacity envelope.

Corollary 10.2 (Non-collapse of the unification). *On the cited source packages, USNRT is a typed host rather than a quotient that identifies theorem-bearing source data. The recovery statement does not assert that arbitrary objects satisfying the bare USNRT axioms must satisfy the source-specific Bellman, pushdown, CCR- k , or exposure theorems.*

10.7 Dependency boundary

The proved USNRT core uses only the source packages and hypotheses explicitly cited above. Later arithmetic refinements, unresolved resource-frontier calculations, or category-specific compiler constructions are specialization questions and are not premises of the exact-realization, protected-obligation, saturation, holonomy, or recovery results in this article.

11 Non-transfer principles and scope

Several tempting implications are invalid without additional hypotheses. They are recorded here because the boundary of a transfer theorem is part of the theorem.

- (N1) **Topology alone does not imply physical cost.** A physical transport representation must first be chosen; a protected readout and capacity law are additionally required for resource consequences.

- (N2) **Chronological shadow loss alone does not imply physical cost.** Lost exact chronology becomes a resource issue only if the lost distinctions remain protected by a future decoder/readout contract.
- (N3) **Geometry alone does not imply physical cost.** A geometric witness needs a protected-realization bridge and then a category-specific capacity envelope.
- (N4) **Pairwise availability does not imply global assembly.** This is measured exactly by the assembly defect δ_{asm} .
- (N5) **A capacity lower bound does not imply a matching construction.** Lower bounds are necessary conditions; exact optimality additionally needs an upper-bound construction.
- (N6) **Abstract product realization does not imply one-stack pushdown product closure.** A Cartesian product may carry two independent physical memories.

These non-transfer principles are not limitations of the framework; they are the mechanism by which it prevents category errors between semantic, informational, topological, and physical statements.

12 Relation to neighboring frameworks

USNRT has substantial classical neighbors, and the distinctions matter for the scope of the contribution. Refinement mappings explicitly relate lower-level state machines to higher-level specifications by state-space maps preserving steps and behaviors [8]; refinement calculi formalize correctness-preserving program and data refinement [9]; abstract interpretation constructs composable hierarchies of semantics through abstraction maps [10]; and coalgebra provides a broad categorical theory of transition and dynamical systems [11]. Accordingly, the existence of an abstract/concrete state map, a semantics-preserving refinement relation, or a categorical transition-system shell is not claimed as new here.

Rewriting theory studies confluence, normal forms, and canonicalization [12, 13]. Fractional covers and zero-error/confusability methods provide classical combinatorial and information-theoretic lower-bound shells [14, 15, 16, 17]. Sheaf-theoretic contextuality identifies local/global incompatibility with global-section obstructions [18]; valuation algebras generalize disagreement across information sources [19]; bundle approaches relate contextuality and holonomy [20]; and resource theories formalize monotones and free operations in operational settings [21]. Thus simplicial compatibility, local/global obstruction, holonomy, fractional covering, and resource monotonicity are likewise not claimed as inventions of USNRT.

The paper-level contribution is the combined typed architecture and its theorem package: (i) exact realization with an optional rather than intrinsic section; (ii) an exact path-outcome criterion and debt formula for fixed-template canonicalization; (iii) protected-obligation compatibility as a separate information layer; (iv) explicit conditional bridges from geometry, chronology, or transported holonomy into that layer; and (v) category-relative transfer saturation that separates missing pairwise compatibility from failure of simultaneous assembly in one physical model. The novelty claim is therefore a claim about this specialized synthesis and the resulting transfer/non-transfer theorems, not about the classical ingredients individually.

To the best of our knowledge after the targeted literature comparison summarized above, no single neighboring framework contains this complete typed package with the same theorem roles. We do not treat that comparison as a historical-priority certificate: later prior art may revise the novelty classification without changing the mathematical statements proved here.

13 External generativity: interval-valued scalar fields

The independent EF-FG scalar-field theory supplies a downstream test of the USNRT architecture [2]. The application is intentionally asymmetric: EF-FG provides native scalar-field theorems, while USNRT supplies a typed interpretation of their common-realization obstruction; the present paper does not claim those scalar-field boundary constructions as a second primary home.

For a finite protected family P , let $K_\sigma(P)$ denote EF-FG feasibility under its native maximal-safe canonical section and $K_{\text{raw}}(P)$ raw common-realization feasibility without imposing that section. The companion paper proves two all-orders boundary results:

$$K_\sigma(P_r) = \partial\Delta^r, \quad K_{\text{raw}}(P_r) = \partial\Delta^r \quad (r \geq 1),$$

using respectively the canonical q-band family and the raw radius-two fan construction.

Proposition 13.1 (USNRT interpretation of the EF-FG boundary families). *Under the protected-obligation interpretation, both EF-FG families are instances of unbounded common-realization obstruction order: every proper protected subfamily is jointly realizable, while the full family is not. The canonical family additionally depends on a chosen section, whereas the raw family does not. No new USNRT axiom is required to state this distinction.*

Proof. The companion EF-FG paper supplies the realizability statements and their proofs. In USNRT, place the protected observer-record obligations as vertices of $\text{Ob}_h(P_r)$. A face is present exactly when one common EF-FG realization satisfies the corresponding protected obligations. Therefore the two cited EF-FG identities translate directly to $\text{Ob}_h(P_r) \cong \partial\Delta^r$. The only difference between the two cases is whether the EF-FG section is part of the realization contract. $\square$

This application is evidence of expressiveness, not a logical premise of the core theory. It also avoids theorem duplication: the scalar constructions, physical lift, and NP-completeness results remain in the EF-FG paper, while USNRT records only the typed consequence relevant to its obligation-complex language.

14 Conclusion

USNRT provides a typed exact-realization language for systems in which semantic evolution is deterministic but physical realization, canonicalization, observation, and resource constraints introduce additional structure. The framework separates three obstruction branches: section factorization/debt, protected information/capacity, and marked topology/transport. Same-model transfer saturation then quantifies how much independently protected information a restrictive realization category can compile simultaneously. The resulting theory is deliberately typed: resource conclusions are never inferred directly from geometry, chronology, or topology without the bridges that make those conclusions meaningful.

The next questions are specialization questions rather than missing core axioms: exact transfer saturation in particular machine categories, sharper category-specific resource frontiers, and additional external realizations. The scalar-field boundary theorems show that the architecture is not confined to pushdown automata, while the recovery results show that the unification does not erase the theorem logic from which it was extracted.

A Theorem-scope and bridge ledger

The purpose of this appendix is to make the logical status of the main theorem families explicit. A statement is *intrinsic/formal* when its conclusion follows from the typed mathematical data

stated in the theorem and does not need a model-specific physical capacity theorem. A statement is *conditional* when an explicit bridge must be supplied. A statement is *category-sensitive* when its numerical physical consequence depends on the admissible realization category or resource evaluation.

A.1 Intrinsic/formal results

The following results belong to the formal USNRT core once their displayed inputs are given:

- (I1) the exact-realization category and the fully faithful identity-realization embedding;
- (I2) the rooted endpoint-factorization theorem and the fixed-template section-defect formula;
- (I3) formation of the protected obligation complex and its finite face-cover invariants;
- (I4) rectangular compatibility-product bounds and the fractional-cover rate identity at the derived-complex level;
- (I5) exact finite transfer-loss telescoping;
- (I6) abstract asynchronous product realization for coordinate products;
- (I7) transfer-saturation arithmetic, pair/assembly defect decomposition, and optimal same-model loss once a compiler category has been fixed.

None of these results, by itself, asserts a stack-height, state-count, alphabet-size, exposure, or Pareto lower bound.

A.2 Conditional bridges

Four bridge types recur throughout the paper.

- (C1) A geometric witness produces protected information only when a witness constructor and protected realization/readout are provided.
- (C2) Exact chronology produces protected information only when a future decoder or equivalent protected obligation distinguishes the exact alternatives. A common shadow is insufficient.
- (C3) Path-outcome multiplicity produces obligation pressure only when the alternatives receive protected repair or continuation labels.
- (C4) Marked loops produce physical orbit information only after a transport action on a carrier is specified.

These hypotheses are not technical decoration. Removing them changes the type of the statement and can make the intended conclusion false.

A.3 Category-sensitive conclusions

A protected-information lower bound becomes a physical resource statement only after a category-specific capacity envelope is supplied. Thus observation capacity, debt/escape envelopes, source-stack exposure, state/stack/alphabet lower bounds, and Pareto exclusions are category-sensitive. The source-stack formula in Section 8.3 is one such specialization; it is not promoted to an invariant of bare semantics.

Likewise, a physical repeated-product interpretation of h_{obl} requires an iterated rectangular realization family. Without that hypothesis, $h_{\text{obl}}(K) = \log \kappa_f(K)$ remains a theorem about the derived protected compatibility complex.

B Four-axis recovery ledger

This appendix gives the recovery audit in the same five-field format for each axis. It is intentionally a verification ledger rather than a second proof of the source theorems.

B.1 Axis I: completion geometry

Source. *Universal Row Complexity of Simplicial Complexes* and *Universal k -Fibers of Optimal Compatibility* [3, 4].

Audited theorem family. Exact Universal Row Complexity and Zero Shell Defect (Theorem 2.2), Universal k -Fiber Theorem, and extension-fiber completeness.

Embedding. Fixed periodic/Bellman system, exact distance, compatibility hierarchy, lower-order signature, higher-order extension fibers, and marked data are placed in the geometric enrichment $\mathcal{G}_\bullet$; pure geometric examples use the identity realization.

Bridge. No physical conclusion is drawn without an additional protected-readout/capacity bridge.

Reconstruction. Forgetting unrelated USNRT layers returns the source system and its theorem-bearing compatibility data; in particular, compatibility fibers are not identified with representation fibers.

B.2 Axis II: exact pushdown normalization

Source. The exact-normalization/general-theory part of [1] and *Confluence and Normalization Choice in Exact Pushdown Lifts* [5].

Audited theorem family. Finite-observation/common-target collapse capacity, the bounded-local three-stage obstruction decomposition, and rooted path-outcome coherence.

Embedding. Semantic dynamics, physical configurations, exact projection, chosen section, bounded observations, protected obligations, and local action/schedule data are carried by the realization and protection layers.

Bridge. Tail-injectivity/noncoalescence and transition closure turn the same-target predecessor family into the complete-conflict protected-obligation specialization; later allocation and local-alignment stages remain source-specific.

Reconstruction. Projection returns the original policy, section, observations, and local action data while preserving the distinction between a capacity screen and exact local realizability.

B.3 Axis III: chronological arithmetic

Source. *Higher-Order Chronological Carry: Connected Interval Blocks, Hessenberg Transport, and Stirling Shadows* [6]; the same theorem family is publicly archived in Chapter 34 of [1].

Audited theorem family. Terminal-block separation, higher-band lower-Hessenberg realization, subdiagonal light cone, normalized forward transport, inverse source reconstruction, stationary parameter lattice, and raw Stirling shadow.

Embedding. Exact connected sources, Hessenberg bands, transport/inverse data, stationary parameters, and shadow maps are stored as separate chronological coordinates in $\mathcal{J}_\bullet$.

Bridge. A common shadow creates protected-information pressure only when a future decoder/readout distinguishes the exact alternatives.

Reconstruction. Forgetting unrelated layers returns exact and shadow data separately, with all source hypotheses for stationarity, divisibility, and transport retained.

B.4 Axis IV: information and resource complexity

Source. *Collapse-Retained Information and Representation-Independent Pushdown Exposure Complexity* [7].

Audited theorem family. Filtered Collapse-Capacity Theorem 2.4 and Source-Stack Exposure Profile Theorem 3.3.

Embedding. The collapse source family, protected contract, observation filtration, confinement predicate, debt/escape counts, and resource parameters are represented in the protected-obligation/resource layers.

Bridge. Theorem 2.4 gives $D + X_h + V(h) \geq N$ under hidden-tail noncoalescence, while Theorem 3.3 gives $D + X_h + q_{\text{src}} \sum_{j=0}^h g_{\text{src}}^j \geq N$ for deterministic top-access pushdown sources. Selecting one source representative per distinct joint protected label sets $N = R$ in the saturation specialization.

Reconstruction. Projection returns the source filtration, exposure depths, debt data, and source-local state/alphabet parameters; exposure is not identified with stack height and a capacity inequality is not identified with a matching compiler.

B.5 Ledger conclusion

For every axis, the reconstruction map returns the audited source package after interpretation, while source-specific hypotheses remain explicit. The ledger therefore certifies source faithfulness; it does not assert that bare USNRT axioms imply any of the four source theorem families.

Statements and declarations

Author contributions. Alp Eren Bütün: conceptualization, methodology, formal analysis, investigation, validation, writing—original draft, and writing—review and editing.

Funding. No funding was received for conducting this study.

Competing interests. The author has no relevant financial or non-financial interests to disclose.

Data and code availability. The paper is theoretical and uses no empirical dataset. Definitions and proofs of the intrinsic USNRT results are contained in the manuscript. The source-recovery statements explicitly cite the external source theorems whose hypotheses and conclusions are being translated; those cited results are not re-proved here. No external dataset or executable code is required for the intrinsic proofs.

Ethics approval and consent. Not applicable.

Consent for publication. Not applicable.

References

- [1] A. E. Bütün, *Deterministic Pushdown Normalization: From $L_{p,q}$ Generalization to General Theory*, Zenodo, 2026. doi:10.5281/zenodo.22801139.
- [2] A. E. Bütün, *Extrema-Frame Foreground-Background Theory: Self-Dual Structure, Exact Common-Realization Semantics, and Ownership Complexity*, companion manuscript, 2026.
- [3] A. E. Bütün, *Universal Row Complexity of Simplicial Complexes: Common-Shell Threshold Systems and Bellman Universality*, Zenodo, 2026. doi:10.5281/zenodo.22709341.
- [4] A. E. Bütün, *Universal k -Fibers of Optimal Compatibility in Periodic Shortest-Path Systems*, Zenodo, 2026. doi:10.5281/zenodo.22709346.

- [5] A. E. Bütün, *Confluence and Normalization Choice in Exact Pushdown Lifts*, Zenodo, 2026. doi:10.5281/zenodo.22709274.
- [6] A. E. Bütün, *Higher-Order Chronological Carry: Connected Interval Blocks, Hessenberg Transport, and Stirling Shadows*, companion manuscript, 2026.
- [7] A. E. Bütün, *Collapse-Retained Information and Representation-Independent Pushdown Exposure Complexity*, Zenodo, 2026. doi:10.5281/zenodo.22709268.
- [8] M. Abadi and L. Lamport, The existence of refinement mappings, *Theoretical Computer Science* 82 (1991), 253–284. doi:10.1016/0304-3975(91)90224-P.
- [9] R.-J. Back and J. von Wright, *Refinement Calculus: A Systematic Introduction*, Springer, New York, 1998. doi:10.1007/978-1-4612-1674-2.
- [10] P. Cousot, Constructive design of a hierarchy of semantics of a transition system by abstract interpretation, *Electronic Notes in Theoretical Computer Science* 6 (1997), 77–102. doi:10.1016/S1571-0661(05)80168-9.
- [11] J. J. M. M. Rutten, Universal coalgebra: a theory of systems, *Theoretical Computer Science* 249 (2000), 3–80. doi:10.1016/S0304-3975(00)00056-6.
- [12] G. Huet, Confluent reductions: abstract properties and applications to term rewriting systems, *Journal of the ACM* 27 (1980), 797–821. doi:10.1145/322217.322230.
- [13] F. Baader and T. Nipkow, *Term Rewriting and All That*, Cambridge University Press, 1998. doi:10.1017/CBO9781139172752.
- [14] L. Lovász, On the ratio of optimal integral and fractional covers, *Discrete Mathematics* 13 (1975), 383–390. doi:10.1016/0012-365X(75)90058-8.
- [15] M. Karchmer, E. Kushilevitz, and N. Nisan, Fractional covers and communication complexity, *SIAM Journal on Discrete Mathematics* 8 (1995), 76–92. doi:10.1137/S0895480192238482.
- [16] H. S. Witsenhausen, The zero-error side information problem and chromatic numbers, *IEEE Transactions on Information Theory* 22 (1976), 592–593. doi:10.1109/TIT.1976.1055607.
- [17] E. R. Scheinerman and D. H. Ullman, *Fractional Graph Theory*, Wiley, 1997.
- [18] S. Abramsky and A. Brandenburger, The sheaf-theoretic structure of non-locality and contextuality, *New Journal of Physics* 13 (2011), 113036. doi:10.1088/1367-2630/13/11/113036.
- [19] S. Abramsky and G. Carù, Non-locality, contextuality and valuation algebras: a general theory of disagreement, *Philosophical Transactions of the Royal Society A* 377 (2019), 20190036. doi:10.1098/rsta.2019.0036.
- [20] S. B. Montanhano, Contextuality in the bundle approach, n-contextuality, and the role of holonomy, arXiv:2105.14132, 2021.
- [21] B. Amaral, Resource theory of contextuality, *Philosophical Transactions of the Royal Society A* 377 (2019), 20190010. doi:10.1098/rsta.2019.0010.